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LLM-based NLG Evaluation: Current Status and Challenges

Mingqi Gao* , **Xinyu Hu*** , **Jie Ruan** , **Xiao Pu** , **Xiaojun Wan**
Peking University
{gaomingqi, huxinyu, wanxiaojun}@pku.edu.cn, {ruanjie, puxiao}@stu.pku.edu.cn

Abstract

Evaluating natural language generation (NLG) is a vital but challenging problem in artificial intelligence. Traditional evaluation metrics mainly capturing content (e.g. n-gram) overlap between system outputs and references are far from satisfactory, and large language models (LLMs) such as ChatGPT have demonstrated great potential in NLG evaluation in recent years. Various automatic evaluation methods based on LLMs have been proposed, including metrics derived from LLMs, prompting LLMs, and fine-tuning LLMs with labeled evaluation data. In this survey, we first give a taxonomy of LLM-based NLG evaluation methods, and discuss their pros and cons, respectively. We also discuss human-LLM collaboration for NLG evaluation. Lastly, we discuss several open problems in this area and point out future research directions.

1 Introduction

Evaluating natural language generation (NLG) is a key but challenging issue. Traditional evaluation metrics like BLEU [Papineni *et al.*, 2002] and ROUGE [Lin, 2004] rely on the n-gram overlap between model outputs and references to measure its quality. They have been criticized for low correlation with human judgments [Sulem *et al.*, 2018], as surface-level matching cannot reliably evaluate text. After the rise of deep learning, model-based evaluation metrics like BERTScore [Zhang *et al.*, 2020] and BARTScore [Yuan *et al.*, 2021] have been continuously proposed and gradually adopted to evaluate the overall quality or various specific aspects of generated outputs (e.g., fluency, coherence, coverage, faithfulness, etc.). Although better than traditional metrics, their performance is still not satisfactory, and their application scope is very limited. For example, BERTScore is reference-based and cannot be used without a reference. With the emergence of large language models (LLMs) like InstructGPT [Ouyang *et al.*, 2022], they have achieved unprecedented effectiveness in following instructions, understanding content, and generating text. This inspired researchers to use LLMs for NLG evaluation. Although this is a research direction that only emerged in 2023, the past year has seen an enormous amount of research work. It is no exaggeration to say that NLG evaluation has been revolutionized by LLMs. This article will review the existing literature and provide suggestions for future research in this field.

*Equal contribution.

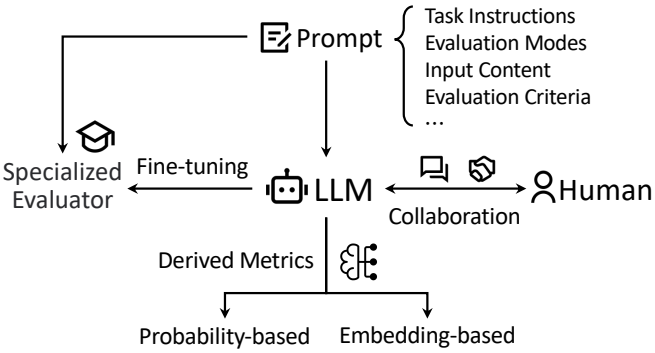

Figure 1: Schematic representation of the four categories of LLM-based NLG evaluation.

This article mainly focuses on research that uses language models with over one billion parameters for NLG evaluation, with necessary references to earlier model-based evaluation metrics like BERTScore. To maintain focus, other types of generation like code generation and tasks involving images are not included in the scope of this article. As shown in Figure 1, according to how we utilize LLMs for NLG evaluation, we categorize the research work into four types:

- **LLM-derived Metrics** (§2): developing/deriving evaluation metrics from embeddings or generation probabilities of LLMs.
- **Prompting LLMs** (§3): directly inquiring existing LLMs via designed prompts which involve different elements for evaluation.
- **Fine-tuning LLMs** (§4): using labeled evaluation data to fine-tune existing LLMs and improving their NLG evaluation capabilities.
- **Human-LLM Collaborative Evaluation** (§5): leveraging distinctive strengths of both human evaluators and LLMs to achieve robust and nuanced evaluations in challenging domains through human-LLM collaboration.

We will review each type of evaluation methods and discuss the pros and cons respectively. Lastly, we will discuss future directions in this area (§6).

2 LLM-derived Metrics

2.1 Overview

Early model-based NLG evaluation methods and metrics, such as BERTScore and BARTScore, were primarily motivated by the capability of traditional pre-trained language models to generate high-quality texts. Recently, the advent of LLMs has prompted some research to adapt these ideas to stronger LLMs. Their inherent, powerful linguistic abilities are derived for direct NLG evaluation, which is expected to result in better performance. Such works can be categorized into two main types: embedding-based metrics and probability-based metrics.

2.2 Embedding-based Metrics

The embedding-based methods, like BERTScore, generally utilize representations of language models and thus compute the semantic similarity between the reference and the target text to evaluate, with different possible ways of implementation. Recent work [ES *et al.*, 2023] similarly obtains embeddings for the target text using the text-embedding-ada-002 model through the OpenAI API. The higher the corresponding similarity, the closer the target text aligns with the relevant requirements, indicating a higher quality.

2.3 Probability-based Metrics

To better utilize the knowledge inherent in language models, probability-based methods like BARTScore formulate text generation evaluation as conditional probability comparison, positing that the better the quality of the target text, the higher the likelihood that models should be able to generate it. Recently, GPTScore [Fu *et al.*, 2023a] has established tailored evaluation templates for each aspect to effectively guide multiple LLMs for NLG evaluation, including GPT3 [Brown *et al.*, 2020], OPT [Zhang *et al.*, 2022], and FLAN [Chung *et al.*, 2022]. The generation probability is calculated under the condition of source input designed with customized prompts and evaluation aspects, which endows the GPTScore with better flexibility in evaluation. And similar methods have also been applied to the hallucination detection of the LLM-generated text [Varshney *et al.*, 2023] with three different attempts for calculating the probability score.

On the other hand, some works leverage the variation in probabilities under changed conditions as the evaluation metric. FFLM [Jia *et al.*, 2023] proposes to evaluate the faithfulness of the target text by calculating a combination of probability changes based on the intuition that the generation probability of a given text segment increases when more consistent information is provided, and vice versa. Similarly, DELTASCORE [Xie *et al.*, 2023] measures the quality of different story aspects according to the likelihood difference between pre- and post-perturbation states with LLMs including GPT-3.5 (text-davinci-003) that provide logits. They believe that the sensitivity to specific perturbations indicates the quality of related aspects, and their experiments demonstrate the effectiveness of their approach.

2.4 Pros and Cons

Traditional NLG evaluation approaches always fall short due to their surface-form similarity when the target text and reference convey the same meaning but use different expressions. In contrast, LLM-derived metrics offer a remedy for the limitation and demonstrate stronger correlations with human judgments benefiting from the evolving modeling techniques. However, the flaws within LLMs can lead to some issues, as introduced in the following:

Robustness. Some research has investigated the robustness of LLM-derived metrics and found that they lack robustness in different attack scenarios. Specifically, [He *et al.*, 2023b] develops a set of stress tests to assess the robustness of various model-based metrics on some common NLG tasks. They show a catalogue of the blind spots and potential errors identified that are not detected by different metrics.

Efficiency. Compared to traditional metrics, LLM-derived evaluation methods are more time-consuming and require more computational resources, especially when adopting LLMs with quite large parameter scales. To address this, [Eddine *et al.*, 2022] proposes an approach to learning a lightweight version of LLM-derived metrics, and some fast LLM inference and serving tools like vLLM [Kwon *et al.*, 2023] have been launched. However, closed-source LLMs often do not make their parameters, representations, or logits public and available, thus making it impossible to apply LLM-derived methods to them.

Fairness. [Sun *et al.*, 2022] assesses the social bias across various metrics for NLG evaluation on six sensitive attributes: race, gender, religion, physical appearance, age, and socioeconomic status. Their findings reveal that model-based metrics carry noticeably more social bias than traditional metrics. Relevant biases can be categorized into two types: intrinsic bias encoded within pre-trained language models and extrinsic bias injected during the computation of similarity. Therefore, current LLM-derived methods may have similar issues.

3 Prompting LLMs

3.1 Overview

LLMs have demonstrated unprecedented instruction understanding and text generation abilities, which broadens researchers' imagination of automatic evaluation for NLG. For a long time, human evaluation has been viewed as the gold standard for NLG evaluation. Recently, some studies claim that LLMs are on par with crowdsourcing annotators in several tasks [Törnberg, 2023; Gilardi *et al.*, 2023; Ostyakova *et al.*, 2023; Cegin *et al.*, 2023]. So, can LLMs simulate or even be an alternative to humans in the human evaluation of NLG? Or, do the practices in human evaluation or other evaluative tasks (e.g. competition race, paper review, etc.) inspire us to better NLG evaluation with LLMs? A large body of research and attempts based on prompting LLMs are guided by these ideas. In these studies, instructions and the text to be evaluated are completely expressed in the prompt given to LLMs, and the evaluation results are generated by them.

Human evaluation typically includes the following elements:

Prompt

You will be given a news article and a summary written for it. Your task is to rate the summary on one metric. Please make sure you read and understand these instructions carefully.

Evaluation Criteria:
Consistency (1-5) - the factual alignment between the summary and the summarized source. A factually consistent summary contains only statements that are entailed by the source document. Annotators were also asked to penalize summaries that contained hallucinated facts.

Source Text:
Paul Merson has restarted his row with Andros Townsend after the Tottenham midfielder was brought on with only seven minutes remaining in his team's 0-0 draw with Burnley on Sunday. ...

Summary:
Paul Merson was brought on with only seven minutes remaining ....

Evaluation Form:
Answer by starting with "Rating:" and then give the explanation of the rating on the next line by "Rationale:".

Response

Rating: 2
Rationale: The summary incorrectly states that Andros Townsend ....

Figure 2: An example of prompting LLMs to evaluate the aspect of consistency of the summary. There are task instructions, evaluation criteria, input content, and evaluation methods in the prompt, as well as the evaluation results, including the rating and explanation generated by LLMs.

- **Evaluation Methods**: The way the preferences of annotators are obtained, such as scoring and comparison.
- **Task Instructions**: How the annotators should read or manipulate different parts to complete the annotation.
- **Input Content**: The target text to be evaluated and other required content. Other required content including source documents, references, and external knowledge is provided as needed.
- **Evaluation Criteria**: The general definition of how good or bad the text to be evaluated is in a particular aspect of quality, e.g. fluency, faithfulness.
- **Role and interaction**: The roles annotators play in the evaluation and the interactions between them.

The focus of the existing research can always be mapped analogously to one or more of these elements, and we organize them according to the elements they address. An example of prompting LLMs is shown in Figure 2.

3.2 Evaluation Methods

Scoring. Scoring is the most commonly used evaluation method in human evaluation for NLG, which is naturally applied to NLG evaluation by prompting LLMs. [Chiang and Lee, 2023a] has conducted relevant studies early, using a Likert scale from 1 to 5 to evaluate story generation and adversarial attacks with InstructGPT [Ouyang *et al.*, 2022] and ChatGPT [1], showing that the evaluation results of LLMs are consistent with expert human evaluators. [Kocmi and Federmann, 2023b] discover GPT-3.5 and GPT-4 achieve the state-of-the-art accuracy of evaluating translation quality compared to human labels, outperforming all the results from the metric shard task of WMT22 [Freitag *et al.*, 2022]. [Wang *et al.*, 2023a] experiment on five datasets across summarization, story generation, and data-to-text, and ChatGPT evaluators with a rating scale from 1 to 5 or 1 to 100 have the state-of-the-art or comparative correlations with human judgments in most settings, compared with prior metrics. Similar conclusions are also observed in open-domain dialogue response generation [Lin and Chen, 2023]. Besides English, [Mendonça *et al.*, 2023] show that ChatGPT with simple rating prompts is a strong evaluator for multilingual dialogue evaluation, surpassing prior metrics based on encoders.

Comparison. Different from absolute scoring, comparison refers to choosing the better of the two. [Luo *et al.*, 2023; Gao *et al.*, 2023] use ChatGPT to compare the factual consistency of two summaries. AuPEL [Wang *et al.*, 2023d] evaluate personalized text generation from three aspects in the form of comparison with the PaLM 2 family [Anil *et al.*, 2023]. According to [Liusie *et al.*, 2023], pairwise comparison is better than scoring when medium-sized LLMs (e.g. FlanT5 [Chung *et al.*, 2022] and Llama2 [Touvron *et al.*, 2023]) are adopted as evaluators.

Ranking. Ranking can be viewed as an extended form of comparison. In comparison, only two examples are involved at a time, whereas in ranking, the order of more than two examples needs to be decided at once. [Ji *et al.*, 2023] use ChatGPT to rank five model-generated responses across several use cases at once, indicating the ranking preferences of ChatGPT align with those of humans to some degree. Similarly, GPTRank is a method to rank summaries in a list-wise manner [Liu *et al.*, 2023c]. Moreover, [Liu *et al.*, 2023b] compare different evaluation methods in LLM-based summarization including scoring, comparison, and ranking, showing that the optimal evaluation method for each backbone LLM may vary.

Boolean QA. Boolean QA requires LLMs to answer "Yes" or "No" to a question. It is adopted more in scenarios where human annotations are binary, such as grammaticality [Hu *et al.*, 2023], faithfulness of summaries and statements [Luo *et al.*, 2023; Gao *et al.*, 2023; ES *et al.*, 2023; Hu *et al.*, 2023], factuality of generated text [Fu *et al.*, 2023b; Guan *et al.*, 2023; Manakul *et al.*, 2023], and answerability of generated questions [Wang *et al.*, 2023f].

Error Analysis. Error Analysis refers to the evaluation of a text by looking for errors that occur in the text according to a set of predefined error categories. Multidimensional Quality Metrics (MQM) [Jain *et al.*, 2023] is an error analysis framework prevalent in machine translation evaluation. According to MQM, [Lu *et al.*, 2023; Kocmi and Federmann, 2023a] use ChatGPT or GPT-4 to automatically detect translation quality error spans. BOOOOKSCORE [Chang *et al.*, 2023], an LLM-based evaluation metric, assesses the coherence of book summaries by identifying eight types of errors.

[1] https://openai.com/blog/chatgpt/

3.3 Task Instructions

In human evaluation, this part usually comes in the form of a task description or evaluation steps. They can also exist at the same time. The task description states the annotation in a more general way, and the evaluation steps, which can be considered as Chain-of-Thought, explicitly describe what to do at each step. In the context of prompting LLMs for NLG evaluation, the demonstrations used in few-shot prompting are also included.

Form and requirements. Several studies from an Eval4NLP 2023 shared task [Leiter *et al.*, 2023] have explored task instructions in various settings. [Kim *et al.*, 2023a] conduct experiments on different templates and lengths of task descriptions and evaluation steps. [Kotonya *et al.*, 2023] generate task instructions with LLMs or improve existing task instructions with LLMs. Additionally, [He *et al.*, 2023a] evaluate generative reasoning using LLMs by asking them first to generate their own answers, and then conduct a quantitative analysis of the text to be evaluated.

Analysis and explanations. LLMs can include analysis or explanation in their evaluations, which is a key point that distinguishes them from previous automatic evaluation metrics. Early explorations into prompting LLMs for NLG evaluation mostly do not examine the impact of whether LLMs are required to analyze and explain on evaluation results. However, [Chiang and Lee, 2023b] explore different types of evaluation instructions in summarization evaluation and dialogue evaluation, finding that explicitly asking large models to provide analysis or explanation achieve higher correlation with human judgments. Besides, the quality of the analysis and explanation generated by LLMs itself requires additional manual evaluation [Leiter *et al.*, 2023]. [Naismith *et al.*, 2023] compare the explanations written by humans and generated by GPT-4 and conduct a simple corpus analysis on the generated explanations.

In-context examples. Similarly to other fields, sometimes demonstrations are needed when prompting LLMs for NLG evaluation. Specifically, [Jain *et al.*, 2023] use only in-context examples as task instructions, relying on LLMs to evaluate the quality of summaries. In scenarios where task descriptions or evaluation steps are included, [Kotonya *et al.*, 2023] compare the performance of LLMs as evaluators in both zero-shot and one-shot settings, finding that one-shot prompting does not bring improvements. Moreover, [Hasanbeig *et al.*, 2023] improve the performance of LLM evaluators by updating the in-context examples iteratively.

3.4 Input Content

The types of input content mainly depend on the evaluation criteria and are relatively fixed. For most task-specific evaluation criteria, such as the faithfulness of a summary [Luo *et al.*, 2023; Gao *et al.*, 2023], the source document is needed in addition to the target text to be evaluated. For task-independent criteria, such as fluency [Hu *et al.*, 2023; Chiang and Lee, 2023b], only the text to be evaluated needs to be provided, though many works also provide the source document [Wang *et al.*, 2023a; Liusie *et al.*, 2023]. Other types of input content can be provided as required by the specific task. [Kocmi and Federmann, 2023b] use two different settings when evaluating machine translation: providing references and not providing references. [Guan *et al.*, 2023] provide relevant facts and context when evaluating whether a text conforms to the facts. Exceptionally, [Shu *et al.*, 2023] add the output of other automatic evaluation metrics to the input of the LLM.

3.5 Evaluation Criteria

The evaluation targeting specific aspects is used in numerous studies of human evaluation for NLG, such as text summarization, story generation, dialogue, and text simplification. Evaluation criteria, i.e., the definitions of aspects are key in this context. Most evaluation criteria in LLM-based evaluation are directly derived from human evaluation. However, a few studies have attempted to let LLMs generate or improve evaluation criteria. [Liu *et al.*, 2023e] use a few human-rated examples as seeds to let LLMs draft some candidate evaluation criteria, and then further filter them based on the performance of LLMs using these criteria on a validation set, to obtain the final evaluation criteria. [Kim *et al.*, 2023c] designed an LLM-based interactive evaluation system, which involves using LLMs to review the evaluation criteria provided by users, including eliminating ambiguities in criteria, merging criteria with overlapping meanings, and decomposing overly broad criteria. Additionally, [Ye *et al.*, 2023a] propose a hierarchical aspect classification system with 12 subcategories, demonstrating that under the proposed fine-grained aspect definitions, human evaluation and LLM-based evaluation are highly correlated. Besides, the chain-of-aspects approach improves LLMs' ability to evaluate on a specific aspect by having LLMs score on some related aspects before generating the final score [Gong and Mao, 2023].

3.6 Role and Interaction

We include in this section the evaluation strategies that either use the same LLMs in different ways or involve different LLMs. The former can be further divided into chain-style and network-style interactions.

Chain-style interaction. Inspired by human evaluators, [Yuan *et al.*, 2024] have LLMs score a batch of examples to be evaluated each time. Specifically, the evaluation process is divided into three stages: analysis, ranking, and scoring. Similar to QA-based evaluation metrics [Durmus *et al.*, 2020], [Fu *et al.*, 2023b] assess the faithfulness of summaries in two stages: treating LLMs as question generators to generate a question from the summary; then having LLMs answer the question using the source document. Differently, when [Hu *et al.*, 2023] use GPT-4 to evaluate the faithfulness of summaries, it first asks GPT-4 to extract event units from the summary, then verifies whether these event units meet the requirements, and finally judges whether the event units are faithful to the source document.

Network-style interaction. Unlike chain-style interactions, network-style interactions involve the dispersion and aggregation of information. In network-style interactions, LLMs on the same layer play similar roles. ChatEval [Chan *et al.*, 2023] is a framework for evaluating content through debates among multiple LLMs, with three communication

strategies designed among the three types of LLMs: One-By-One, Simultaneous-Talk, and Simultaneous-Talk-with-Summarizer. [Zhang *et al.*, 2023b] find that under certain conditions, widening and deepening the network of LLMs can better align its evaluation with human judgments. [Saha *et al.*, 2023] propose a branch-solve-merge strategy, assigning LLMs the roles of decomposing problems, solving them, and aggregating answers, thereby improving the accuracy and reliability of evaluations. [Wu *et al.*, 2023] assume that different people such as politicians and the general public have different concerns about the quality of news summaries, use LLMs to play different roles in evaluation accordingly, and aggregate the results finally.

Different LLMs. Different from having the same LLM play different roles, some research has used different LLMs (such as GPT-4 and Claude) in their studies. In pairwise comparisons, previous work mostly used a single LLM as the evaluator, which may not be fair. In light of this, [Bai *et al.*, 2023] design a decentralized Peer-examination method, using different LLMs as evaluators and then aggregating the results. Further, [Li *et al.*, 2023c] let different LLMs serve as evaluators in pairwise comparisons and then have them go through a round of discussion to reach the final result. Additionally, [Cohen *et al.*, 2023] evaluate the factuality of texts through the interaction of two LLMs, where the LLM that generated the text acts as the examinee and the other LLM as the examiner.

3.7 Pros and Cons

The benefits of prompting LLMs for NLG evaluation are exciting. First, for the first time, people can express evaluation criteria and evaluation methods in natural language within the prompts given to LLMs, providing great flexibility. Where previously people needed to design specific evaluation metrics for different NLG tasks or even different aspects of a single task, now they only need to modify the prompts for LLMs. Secondly, surprisingly, LLMs have the ability to generate explanations while assessing texts, making this approach somewhat interpretable. Furthermore, in many NLG task, prompting LLMs for evaluation has achieved state-of-the-art correlations with human judgments.

However, as many studies have pointed out, this type of approach still has many limitations. [Wang *et al.*, 2023b] note that when using ChatGPT and GPT-4 for pairwise comparisons, the order of the two texts can affect the evaluation results, which is known as position bias. To alleviate this issue, [Li *et al.*, 2023d] propose a strategy of splitting, aligning, and then merging the two texts to be evaluated into the prompt. Also, LLM evaluators tend to favor longer, more verbose responses [Zheng *et al.*, 2023] and responses generated by themselves [Liu *et al.*, 2023a]. [Wu and Aji, 2023] show that compared to answers that are too short or grammatically incorrect, answers with factual errors are considered better by LLMs. [Liu *et al.*, 2023d] demonstrate through adversarial meta-evaluation that LLMs without references are not suitable for evaluating dialogue responses in closed-ended scenarios: they tend to score highly on responses that conflict with the facts in the dialogue history. [Zhang *et al.*, 2023a] also present the robustness issues of LLMs in dialogue evaluation through adversarial perturbations. [Shen *et al.*, 2023] indicate that LLM evaluators have a lower correlation with human assessments when scoring high-quality summaries. In addition, [Hada *et al.*, 2023] state that LLM-based evaluators have a bias towards high scores, especially in non-Latin languages like Chinese and Japanese. Beyond these shortcomings of performance, both ChatGPT and GPT-4 are proprietary models, and their opacity could lead to irreproducible evaluation results.

4 Fine-tuning LLMs

4.1 Overview

As mentioned above, despite the exciting performance of prompting LLMs like ChatGPT and GPT-4 for NLG evaluation, several shortcomings in practice are inevitable, such as high costs, possibly irreproducible results, and potential biases in LLMs. In response, recent research has shifted towards fine-tuning smaller, open-source LLMs specifically for evaluation purposes, aiming to achieve performance close to GPT-4 in NLG evaluation. Representative works of this type include PandaLM [Wang *et al.*, 2023e], Prometheus [Kim *et al.*, 2023b], Shepherd [Wang *et al.*, 2023c], TIGERScore [Jiang *et al.*, 2023], INSTRUCTSCORE [Xu *et al.*, 2023], Auto-J [Li *et al.*, 2023a], CritiqueLLM [Ke *et al.*, 2023] and JudgeLM [Zhu *et al.*, 2023]. Their main ideas are similar, involving the elaborate construction of high-quality evaluation data, followed by accordingly fine-tuning open-source base LLMs. Nevertheless, there are certain discrepancies in the designs across different works, such as the usage of references and evaluation criteria. We have summarized the key different components of these methods in Table 1 for comparison, which will be elaborated on next.

4.2 Data Construction

Diverse data with high-quality annotations is crucial for the fine-tuning of evaluation models, which mainly involves task scenarios, inputs, target texts to evaluate, and evaluation results. Early NLG evaluation research primarily focused on conventional NLG tasks, such as summarization and dialogue generation. Thus, the task scenarios, inputs, and target texts refer to the corresponding NLP task, source inputs of the task, and outputs generated by specialized systems based on task requirements, respectively. And mainstream datasets for these tasks predominantly employ human annotators to provide evaluation results, which are often considered reliable.

With the recent rise of LLMs, the spectrum of NLG tasks has been broadened to scenarios of instruction and response that are more aligned with human needs. Traditional tasks like summarization with corresponding source inputs can be viewed as kinds of instructions and requirements. Meanwhile, responses generated by various general LLMs generally serve as the target texts now and require more flexible evaluation so that the performance of different LLMs can be compared, promoting further developments. Therefore, to keep pace with the current advancement of modeling techniques, most evaluation methods have adopted the similar instruction-response scenario.

Method	Data Construction			Evaluation Method			Base LLM	Reference Required
	Instruction Source	Annotator	Scale	Result Mode	Details	Specific Criteria		
PandaLM	Alpaca 52K	GPT-3.5	300K	Comparison	Reason & Reference	Unified	LLaMA 7B	No
Prometheus	GPT-4 Construction	GPT-4	100K	Scoring	Reason	Explicit	LLaMA-2-Chat 7B & 13B	Yes
Shepherd	Community Critique Data & 9 NLP Tasks Data	Human	1317	Overall Judgement	Error Identifying & Refinement	Unified	LLaMA 7B	No
TIGERScore	23 Distinctive Text Generation Datasets	GPT-4	48K	MQM	Error Analysis	Implicit	LLaMA-2 7B & 13B	No
INSTRUCTSCORE	GPT-4 Construction	GPT-4	40K	MQM	Error Analysis	Implicit	LLaMA 7B	Yes
AUTO-J	Real-world User Queries from Preference Datasets	GPT-4	4396	Scoring & Comparison	Reason	Implicit	LLaMA-2-Chat 13B	No
CritiqueLLM	AlignBench & ChatGPT Augmentation	GPT-4	9332	Scoring	Reason	Unified	ChatGLM-2 6B, 12B & 66B	Flexible
JudgeLM	GPT4All-LAION, ShareGPT Alpaca-GPT4 & Dolly-15K	GPT-4	100K	Scoring & Comparison	Reason	Unified	Vicuna 7B, 13B & 33B	Flexible

Table 1: Comparison of the different key components among the representative methods of fine-tuning LLMs.

The primary differences in these works actually lie in the construction of instructions, with the purpose of improving either diversity or reliability for the better generalization ability of the fine-tuned model. PandaLM and JudgeLM entirely sample from common instruction datasets, such as Alpaca 52K, while CritiqueLLM adopts small-scale sampling followed by ChatGPT augmentation. In contrast, Prometheus and INSTRUCTSCORE rely on GPT-4 to generate all the instructions based on seed data, whereas Auto-J and Shepherd use real-world data. Moreover, since large-scale human annotation is impractical, most works utilize GPT-4 as the powerful annotator, except for PandaLM and Shepherd, which use GPT-3.5 and human annotation on small-scale community data, respectively. During the construction, they basically all design detailed prompts or guidance and apply heuristic filtering strategies and post-processing methods to mitigate noise. Overall, despite the possible higher quality of human annotation, the corresponding drawback is the difficulty in constructing large-scale datasets, which in turn may hinder adequate model training, while using LLMs for construction is the opposite situation.

4.3 Evaluation Method

As with prompting LLMs, the evaluation methods adopted in these works are highly diversified, involving different evaluation criteria, result modes, and usages of the reference. Given that current instruction-response scenarios encompass different types of tasks, it is unsuitable to specify unified evaluation criteria as in traditional NLG tasks. However, some works still do it this way, while some other methods let LLM annotators adaptively and implicitly reflect the required criteria in their evaluations, like PandaLM, TIGERScore, and AUTO-J. In particular, AUTO-J has meticulously crafted 332 evaluation criteria, matched to different tasks. Furthermore, Prometheus explicitly incorporates evaluation criteria into the inputs of the model, expecting flexible evaluation based on various customized criteria.

More details about the evaluation methods are shown in Table 1. All the works require models to provide detailed information, such as reasons for their evaluation results. And the MQM mode can achieve more informative error analysis, offering stronger interpretability. Moreover, some works do not necessarily require references and then have greater value in practice. And a more optimal method is to concurrently support both reference-based and reference-free evaluations as JudgeLM and CritiqueLLM.

4.4 Fine-tuning Implementation

The fine-tuning process is uniformly implemented by different works on their selected open-source LLMs, like LLaMA, and respective constructed data, with some targeted settings. Specifically, Prometheus maintains balanced data distributions during fine-tuning, including the length and label. JudgeLM eliminates potential biases by randomly swapping sample pairs to be compared and randomly removing references. INSTRUCTSCORE utilizes GPT-4 to provide error annotations for the intermediate outputs of the fine-tuned model for further supervised reinforcement. And based on some preliminary experiments and manual analysis, TIGERScore determines appropriate ratios of different types of data during fine-tuning, which are claimed to be crucial by them. Moreover, CritiqueLLM implements separately, with and without references, and explores the effects of data and model scale. Compared to the vanilla fine-tuning setting, these methods have improved the efficiency of model training and the robustness of evaluations.

4.5 Pros and Cons

Those shortcomings of prompting LLM for NLG evaluation can be significantly alleviated due to the customized implementation of data construction and model fine-tuning here. For instance, most fine-tuned models range between 7B and 13B in the scale of parameters, facilitating low-cost inference use and good reproducibility, with performance close to GPT4 in NLG evaluation. And specific measures can be

adopted to prevent related biases found in GPT4 during different stages. Furthermore, this type of approach allows for continuous iteration and improvement of the model to address potential deficiencies or emerging issues discovered in future applications.

However, some biases associated with GPT4 may still persist, as the data construction of most methods employs GPT4 for critical evaluation annotation. On the other hand, the base open-source LLMs selected by existing works are primarily the series of LLaMA. With the rapid updates and improvements of open-source large models recently, it adheres to the intuition that employing a more powerful base LLM should lead to better evaluation performance. However, this means repetitive fine-tuning processes and computational expenses from scratch since directly migrating existing fine-tuned models to the new base LLM is difficult.

Additionally, although many existing methods aspire for more flexible and comprehensive evaluation through fine-tuning, demanding excessive evaluation settings may ultimately lead to poor performance or failure in model training, as found by AUTO-J and CritiqueLLM on criteria and references, respectively. However, there are some disagreements here since Prometheus and JudgeLM show different results. Moreover, considering the different evaluation settings in existing works, it is challenging to conduct a horizontal comparison among them. These issues require further exploration in future research.

5 Human-LLM Collaborative Evaluation

5.1 Overview

While LLMs demonstrate robust evaluation capabilities, there exists a need for further enhancement in terms of their reliability, particularly in establishing a stronger correlation with human evaluation outcomes. Although human evaluation is the gold-standard evaluation approach in NLG, it is recognized for its associated high costs and susceptibility to subjective biases [van der Lee *et al.*, 2021; Deriu *et al.*, 2021; Li *et al.*, 2023b]. The robust and comprehensive capabilities exhibited by LLMs underscore considerable potential for the development of collaborative evaluation methodologies that integrate both human and LLMs. In recent investigations, researchers have initiated the exploration of collaborative evaluation paradigms, which include traditional NLG evaluation methods such as scoring and explaining [Zhang *et al.*, 2021; Li *et al.*, 2023b], broader evaluation methods such as testing and debugging [Ribeiro and Lundberg, 2022], and auditing NLG models to ensure fairness [Rastogi *et al.*, 2023]. Furthermore, scholars [Saunders *et al.*, 2022] are actively engaging in efforts to address the intricate challenge of scalable oversight [Amodei *et al.*, 2016] through the collaboration of humans and LLMs. The objective is to devise strategies for effectively evaluating models in tasks that pose inherent difficulties for human assessors. This collaborative approach seeks to leverage the distinctive strengths of both human evaluators and sophisticated language models to achieve robust and nuanced evaluations in challenging domains.

5.2 Scoring and Explaining

Automated evaluation frequently exhibits a limited correlation with human judgments, while human evaluation, though reliable, is labor-intensive. [Zhang *et al.*, 2021] present a human-machine collaborative framework (HMCEval) which conceptualizes dialogue evaluation as a sample assignment problem to ensure the reliability of evaluation outcomes while minimizing human effort and achieves 99% accuracy with half human effort. Recently, LLMs have emerged as a cost-effective alternative to human evaluations. However, both humans and LLMs have limitations, including inherent subjectivity and unreliable judgments, especially in open-ended tasks with diverse requirements.

To address challenges associated with inconsistent evaluation criteria in open-ended tasks and explore synergy between humans and LLM-based evaluators, [Li *et al.*, 2023b] proposes a Collaborative Evaluation pipeline (COEVAL), which involves designing a checklist of task-specific criteria and conducting detailed evaluations where LLMs generate initial ideation and humans engage in scrutiny. Depending solely on score predictions is insufficient for ensuring reliable evaluation and error detection, particularly when specific criteria demand nuanced analysis beyond straightforward scoring. Building upon recent developments in explainable NLP ([Yin and Neubig, 2022; Jung *et al.*, 2022; Ribeiro and Lundberg, 2022; Ye *et al.*, 2023b]), COEVAL is assigned the additional task of generating explanations to elucidate evaluation outcomes to facilitate a trustworthy collaborative evaluation process. Results indicate COEVAL effectively evaluates lengthy texts by utilizing LLMs, saving significant time and reducing human evaluation outliers. Despite the involvement of LLMs, human scrutiny remains essential, contributing to the revision of around 20% of LLM evaluation scores for enhanced reliability.

5.3 Broader Evaluation Tasks

The broader evaluation of NLG models involves testing and debugging the models. Current methods often rely on highly variable human creativity and extensive manual effort or are limited to addressing a very specific class of bugs. [Ribeiro and Lundberg, 2022] introduce AdaTest, a process that uses LLMs in collaboration with human feedback to automatically generate unit tests that highlight bugs in a target model, which proves to make users 5-10 times more effective at identifying bugs and assists users in effectively fixing bugs without introducing new ones. Moreover, LLMs have shown biases and irresponsible behavior, necessitating thorough auditing before deployment [Blodgett *et al.*, 2020; Jones and Steinhardt, 2022]. AdaTest++ [Rastogi *et al.*, 2023] draw on insights from literature on human-AI collaboration and sensemaking, and engage with research experts in safe and fair AI, which emphasizes the importance of sensemaking and effective communication between humans and AI to capitalize on their complementary strengths in collaborative auditing. AdaTest++ successfully leverages human strengths, such as schematization and hypothesis testing. Moreover, users identified a range of failure modes across 26 different topics in issues that were revealed in formal audits and those that were

previously under-reported. Additionally, ensuring trustworthiness in LLMs for challenging tasks [Chen *et al.*, 2021; Nakano *et al.*, 2021; Li *et al.*, 2022; Menick *et al.*, 2022] poses a crucial challenge. Scalable oversight [Amodei *et al.*, 2016] aims to effectively evaluate models on tasks challenging for humans and suggests the use of AI for assistance. [Saunders *et al.*, 2022] explored providing critiques of model outputs as a form of assistance, demonstrating that model-generated critiques assist humans in identifying overlooked flaws.

5.4 Pros and Cons

The advantages of human-AI collaborative evaluation lie in achieving a balance between efficiency and cost, as demonstrated by COEVAL [Li *et al.*, 2023b] achieving this equilibrium. Additionally, there are complementary strengths between humans and AI. For instance, AdaTest++ [Rastogi *et al.*, 2023] empowers users to consistently utilize their strengths throughout the auditing process, benefiting significantly from LLM. Users who generate the most topics heavily rely on LLM suggestions while employing their contextual reasoning and semantic understanding to update their mental models vigilantly and identify model failures.

However, there are drawbacks. The evaluation results of LLMs may be sensitive to the formats used to query the model and might require additional support for prompt writing [Li *et al.*, 2023b; Rastogi *et al.*, 2023]. Furthermore, the current capability to assess confidence levels is not strong enough, making it challenging to determine when to trust the LLM. Furthermore, certain level of human supervision is still necessary, making it less convenient and cost-effective compared to fully automated evaluation.

6 Conclusions and Future Trends

Through the above review of studies on NLG evaluation based on LLMs, we find that these four categories of approaches have their respective strengths and weaknesses, and most of the existing work is concentrated on prompting LLMs. In view of this, we offer some suggestions for future directions in this field.

Unified benchmarks for LLM-based NLG evaluation approaches. As mentioned above, each of the studies that fine-tuned LLMs to construct specialized evaluation models uses different settings and data during testing, making them incomparable. In the research on prompting LLMs for NLG evaluation, there are some publicly available human judgments on the same NLG task, such as SummEval for summarization. However, the existing human judgments have many problems. Firstly, most of the existing data only involve one type of NLG task and a single human evaluation method (e.g., scoring), making it difficult to evaluate LLMs' performance on different tasks, as well as using different evaluation methods on the same task. Secondly, many of the texts in these human judgments are generated by outdated models (such as Pointer Network) and do not include texts generated by more advanced LLMs. Lastly, many human evaluation datasets are too small in scale. There is an urgent need for large-scale, high-quality human evaluation data covering various NLG tasks and evaluation methods as a benchmark.

NLG evaluation for low-resource languages and new task scenarios. Almost all existing research focuses on English data. However, it is doubtful whether LLMs have similar levels of NLG evaluation capability for texts in other languages, especially low-resource languages. As [Zhang *et al.*, 2023a] points out, we should be more cautious about using LLMs to evaluate texts in non-Latin languages. Additionally, existing research mainly focuses on more traditional NLG tasks such as translation, summarization, and dialogue. However, there are many new scenarios in reality with different requirements and evaluation criteria. Research on low-resource languages and new task scenarios will provide a more comprehensive understanding of LLMs' evaluation capabilities.

Diverse forms of human-LLM collaborative NLG evaluation. According to the literature reviewed above, there is little research on collaborative evaluation between humans and LLMs. Neither humans nor LLMs are perfect, and each has its strengths. Since the ultimate goal of NLG research is to evaluate text quality more accurately and efficiently, we believe that collaboration between humans and LLMs can achieve better results than pure human evaluation or automatic evaluation. In the collaboration between humans and LLMs, technologies in the field of human-computer interaction may bring new implementation methods to the collaboration. In addition, what roles humans and LLMs should play in the evaluation and how they can better complement each other are still worth researching.

References

[Amodei *et al.*, 2016] Dario Amodei, Chris Olah, Jacob Steinhardt, Paul F. Christiano, John Schulman, and Dan Mané. Concrete problems in AI safety. *CoRR*, abs/1606.06565, 2016.

[Anil *et al.*, 2023] Rohan Anil, Andrew M. Dai, Orhan Firat, Melvin Johnson, Dmitry Lepikhin, Alexandre Passos, Siamak Shakeri, Emanuel Taropa, Paige Bailey, Zhifeng Chen, Eric Chu, Jonathan H. Clark, Laurent El Shafey, Yanping Huang, Kathy Meier-Hellstern, Gaurav Mishra, Erica Moreira, Mark Omernick, Kevin Robinson, Sebastian Ruder, Yi Tay, Kefan Xiao, Yuanzhong Xu, Yujing Zhang, Gustavo Hernandez Abrego, Junwhan Ahn, Jacob Austin, Paul Barham, Jan Botha, James Bradbury, Siddhartha Brahma, Kevin Brooks, Michele Catasta, Yong Cheng, Colin Cherry, Christopher A. Choquette-Choo, Aakanksha Chowdhery, Clément Crepy, Shachi Dave, Mostafa Dehghani, Sunipa Dev, Jacob Devlin, Mark Díaz, Nan Du, Ethan Dyer, Vlad Feinberg, Fangxiaoyu Feng, Vlad Fienber, Markus Freitag, Xavier Garcia, Sebastian Gehrmann, Lucas Gonzalez, Guy Gur-Ari, Steven Hand, Hadi Hashemi, Le Hou, Joshua Howland, Andrea Hu, Jeffrey Hui, Jeremy Hurwitz, Michael Isard, Abe Ittycheriah, Matthew Jagielski, Wenhao Jia, Kathleen Kenealy, Maxim Krikun, Sneha Kudugunta, Chang Lan, Katherine Lee, Benjamin Lee, Eric Li, Music Li, Wei Li, YaGuang Li, Jian Li, Hyeontaek Lim, Hanzhao Lin, Zhongtao Liu, Frederick Liu, Marcello Maggioni, Aroma Mahendru, Joshua Maynez, Vedant Misra, Maysam Moussalem, Zachary Nado, John Nham, Eric Ni, Andrew Nys-

trom, Alicia Parrish, Marie Pellat, Martin Polacek, Alex Polozov, Reiner Pope, Siyuan Qiao, Emily Reif, Bryan Richter, Parker Riley, Alex Castro Ros, Aurko Roy, Brennan Saeta, Rajkumar Samuel, Renee Shelby, Ambrose Slone, Daniel Smilkov, David R. So, Daniel Sohn, Simon Tokumine, Dasha Valter, Vijay Vasudevan, Kiran Vodrahalli, Xuezhi Wang, Pidong Wang, Zirui Wang, Tao Wang, John Wieting, Yuhuai Wu, Kelvin Xu, Yunhan Xu, Linting Xue, Pengcheng Yin, Jiahui Yu, Qiao Zhang, Steven Zheng, Ce Zheng, Weikang Zhou, Denny Zhou, Slav Petrov, and Yonghui Wu. Palm 2 technical report. *Computing Research Repository*, arxiv:2305.10403, 2023.

[Bai *et al.*, 2023] Yushi Bai, Jiahao Ying, Yixin Cao, Xin Lv, Yuze He, Xiaozhi Wang, Jifan Yu, Kaisheng Zeng, Yijia Xiao, Haozhe Lyu, Jiayin Zhang, Juanzi Li, and Lei Hou. Benchmarking foundation models with language-model-as-an-examiner. *CoRR*, abs/2306.04181, 2023.

[Blodgett *et al.*, 2020] Su Lin Blodgett, Solon Barocas, Hal Daumé III, and Hanna M. Wallach. Language (technology) is power: A critical survey of "bias" in NLP. In *ACL*, 2020.

[Brown *et al.*, 2020] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. Language models are few-shot learners. In *NeurIPS*, 2020.

[Cegin *et al.*, 2023] Jan Cegin, Jakub Simko, and Peter Brusilovsky. ChatGPT to replace crowdsourcing of paraphrases for intent classification: Higher diversity and comparable model robustness. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 1889–1905, Singapore, December 2023. Association for Computational Linguistics.

[Chan *et al.*, 2023] Chi-Min Chan, Weize Chen, Yusheng Su, Jianxuan Yu, Wei Xue, Shanghang Zhang, Jie Fu, and Zhiyuan Liu. Chateval: Towards better llm-based evaluators through multi-agent debate. *CoRR*, abs/2308.07201, 2023.

[Chang *et al.*, 2023] Yapei Chang, Kyle Lo, Tanya Goyal, and Mohit Iyyer. Booookscore: A systematic exploration of book-length summarization in the era of llms. *CoRR*, abs/2310.00785, 2023.

[Chen *et al.*, 2021] Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Pondé de Oliveira Pinto, Jared Kaplan, Harrison Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, Alex Ray, Raul Puri, Gretchen Krueger, Michael Petrov, Heidy Khlaaf, Girish Sastry, Pamela Mishkin, Brooke Chan, Scott Gray, Nick Ryder, Mikhail Pavlov, Alethea Power, Lukasz Kaiser, Mohammad Bavarian, Clemens Winter, Philippe Tillet, Felipe Petroski Such, Dave Cummings, Matthias Plappert, Fotios Chantzis, Elizabeth Barnes, Ariel Herbert-Voss, William Hebgen Guss, Alex Nichol, Alex Paino, Nikolas Tezak, Jie Tang, Igor Babuschkin, Suchir Balaji, Shantanu Jain, William Saunders, Christopher Hesse, Andrew N. Carr, Jan Leike, Joshua Achiam, Vedant Misra, Evan Morikawa, Alec Radford, Matthew Knight, Miles Brundage, Mira Murati, Katie Mayer, Peter Welinder, Bob McGrew, Dario Amodei, Sam McCandlish, Ilya Sutskever, and Wojciech Zaremba. Evaluating large language models trained on code. *CoRR*, abs/2107.03374, 2021.

[Chiang and Lee, 2023a] David Cheng-Han Chiang and Hung-yi Lee. Can large language models be an alternative to human evaluations? In *ACL (1)*, 2023.

[Chiang and Lee, 2023b] David Cheng-Han Chiang and Hung-yi Lee. A closer look into using large language models for automatic evaluation. In *EMNLP (Findings)*, 2023.

[Chung *et al.*, 2022] Hyung Won Chung, Le Hou, Shayne Longpre, Barret Zoph, Yi Tay, William Fedus, Eric Li, Xuezhi Wang, Mostafa Dehghani, Siddhartha Brahma, Albert Webson, Shixiang Shane Gu, Zhuyun Dai, Mirac Suzgun, Xinyun Chen, Aakanksha Chowdhery, Sharan Narang, Gaurav Mishra, Adams Yu, Vincent Y. Zhao, Yanping Huang, Andrew M. Dai, Hongkun Yu, Slav Petrov, Ed H. Chi, Jeff Dean, Jacob Devlin, Adam Roberts, Denny Zhou, Quoc V. Le, and Jason Wei. Scaling instruction-finetuned language models. *CoRR*, abs/2210.11416, 2022.

[Cohen *et al.*, 2023] Roi Cohen, May Hamri, Mor Geva, and Amir Globerson. LM vs LM: detecting factual errors via cross examination. In *EMNLP*, 2023.

[Deriu *et al.*, 2021] Jan Deriu, Álvaro Rodrigo, Arantxa Otegi, Guillermo Echegoyen, Sophie Rosset, Eneko Agirre, and Mark Cieliebak. Survey on evaluation methods for dialogue systems. *Artif. Intell. Rev.*, 54(1):755–810, 2021.

[Durmus *et al.*, 2020] Esin Durmus, He He, and Mona T. Diab. FEQA: A question answering evaluation framework for faithfulness assessment in abstractive summarization. In *ACL*, 2020.

[Eddine *et al.*, 2022] Moussa Kamal Eddine, Guokan Shang, Antoine J.-P. Tixier, and Michalis Vazirgiannis. Frugalscore: Learning cheaper, lighter and faster evaluation metrics for automatic text generation. In *ACL (1)*, 2022.

[ES *et al.*, 2023] Shahul ES, Jithin James, Luis Espinosa Anke, and Steven Schockaert. RAGAS: automated evaluation of retrieval augmented generation. *CoRR*, abs/2309.15217, 2023.

[Freitag *et al.*, 2022] Markus Freitag, Ricardo Rei, Nitika Mathur, Chi-kiu Lo, Craig Stewart, Eleftherios Avramidis, Tom Kocmi, George F. Foster, Alon Lavie, and André F. T. Martins. Results of WMT22 metrics shared task: Stop using BLEU - neural metrics are better and more robust. In *WMT*, 2022.

[Fu *et al.*, 2023a] Jinlan Fu, See-Kiong Ng, Zhengbao Jiang, and Pengfei Liu. Gptscore: Evaluate as you desire. *CoRR*, abs/2302.04166, 2023.

[Fu *et al.*, 2023b] Xue-Yong Fu, Md. Tahmid Rahman Laskar, Cheng Chen, and Shashi Bhushan TN. Are large language models reliable judges? A study on the factuality evaluation capabilities of llms. *CoRR*, abs/2311.00681, 2023.

[Gao *et al.*, 2023] Mingqi Gao, Jie Ruan, Renliang Sun, Xunjian Yin, Shiping Yang, and Xiaojun Wan. Human-like summarization evaluation with chatgpt. *CoRR*, abs/2304.02554, 2023.

[Gilardi *et al.*, 2023] Fabrizio Gilardi, Meysam Alizadeh, and Maël Kubli. Chatgpt outperforms crowd-workers for text-annotation tasks. *CoRR*, abs/2303.15056, 2023.

[Gong and Mao, 2023] Peiyuan Gong and Jiaxin Mao. Coascore: Chain-of-aspects prompting for NLG evaluation. *CoRR*, abs/2312.10355, 2023.

[Guan *et al.*, 2023] Jian Guan, Jesse Dodge, David Wadden, Minlie Huang, and Hao Peng. Language models hallucinate, but may excel at fact verification. *CoRR*, abs/2310.14564, 2023.

[Hada *et al.*, 2023] Rishav Hada, Varun Gumma, Adrian de Wynter, Harshita Diddee, Mohamed Ahmed, Monojit Choudhury, Kalika Bali, and Sunayana Sitaram. Are large language model-based evaluators the solution to scaling up multilingual evaluation? *CoRR*, abs/2309.07462, 2023.

[Hasanbeig *et al.*, 2023] Hosein Hasanbeig, Hiteshi Sharma, Leo Betthauser, Felipe Vieira Frujeri, and Ida Momennejad. ALLURE: auditing and improving llm-based evaluation of text using iterative in-context-learning. *CoRR*, abs/2309.13701, 2023.

[He *et al.*, 2023a] Hangfeng He, Hongming Zhang, and Dan Roth. Socreval: Large language models with the socratic method for reference-free reasoning evaluation. *CoRR*, abs/2310.00074, 2023.

[He *et al.*, 2023b] Tianxing He, Jingyu Zhang, Tianle Wang, Sachin Kumar, Kyunghyun Cho, James R. Glass, and Yulia Tsvetkov. On the blind spots of model-based evaluation metrics for text generation. In *ACL (1)*, 2023.

[Hu *et al.*, 2023] Yebowen Hu, Kaiqiang Song, Sangwoo Cho, Xiaoyang Wang, Hassan Foroosh, and Fei Liu. Decipherpref: Analyzing influential factors in human preference judgments via GPT-4. In *EMNLP*, 2023.

[Jain *et al.*, 2023] Sameer Jain, Vaishakh Keshava, Swarnashree Mysore Sathyendra, Patrick Fernandes, Pengfei Liu, Graham Neubig, and Chunting Zhou. Multi-dimensional evaluation of text summarization with in-context learning. In *ACL (Findings)*, 2023.

[Ji *et al.*, 2023] Yunjie Ji, Yan Gong, Yiping Peng, Chao Ni, Peiyan Sun, Dongyu Pan, Baochang Ma, and Xiangang Li. Exploring chatgpt's ability to rank content: A preliminary study on consistency with human preferences. *CoRR*, abs/2303.07610, 2023.

[Jia *et al.*, 2023] Qi Jia, Siyu Ren, Yizhu Liu, and Kenny Q. Zhu. Zero-shot faithfulness evaluation for text summarization with foundation language model. In *EMNLP*, 2023.

[Jiang *et al.*, 2023] Dongfu Jiang, Yishan Li, Ge Zhang, Wenhao Huang, Bill Yuchen Lin, and Wenhu Chen. Tigerscore: Towards building explainable metric for all text generation tasks. *CoRR*, abs/2310.00752, 2023.

[Jones and Steinhardt, 2022] Erik Jones and Jacob Steinhardt. Capturing failures of large language models via human cognitive biases. In *NeurIPS*, 2022.

[Jung *et al.*, 2022] Jaehun Jung, Lianhui Qin, Sean Welleck, Faeze Brahman, Chandra Bhagavatula, Ronan Le Bras, and Yejin Choi. Maieutic prompting: Logically consistent reasoning with recursive explanations. In *EMNLP*, 2022.

[Ke *et al.*, 2023] Pei Ke, Bosi Wen, Zhuoer Feng, Xiao Liu, Xuanyu Lei, Jiale Cheng, Shengyuan Wang, Aohan Zeng, Yuxiao Dong, Hongning Wang, Jie Tang, and Minlie Huang. Critiquellm: Scaling llm-as-critic for effective and explainable evaluation of large language model generation. *CoRR*, abs/2311.18702, 2023.

[Kim *et al.*, 2023a] Joonghoon Kim, Saeran Park, Kiyoon Jeong, Sangmin Lee, Seung Hun Han, Jiyoon Lee, and Pilsung Kang. Which is better? exploring prompting strategy for llm-based metrics. *CoRR*, abs/2311.03754, 2023.

[Kim *et al.*, 2023b] Seungone Kim, Jamin Shin, Yejin Cho, Joel Jang, Shayne Longpre, Hwaran Lee, Sangdoo Yun, Seongjin Shin, Sungdong Kim, James Thorne, and Minjoon Seo. Prometheus: Inducing fine-grained evaluation capability in language models. *CoRR*, abs/2310.08491, 2023.

[Kim *et al.*, 2023c] Tae Soo Kim, Yoonjoo Lee, Jamin Shin, Young-Ho Kim, and Juho Kim. Evallm: Interactive evaluation of large language model prompts on user-defined criteria. *CoRR*, abs/2309.13633, 2023.

[Kocmi and Federmann, 2023a] Tom Kocmi and Christian Federmann. GEMBA-MQM: detecting translation quality error spans with GPT-4. In *WMT*, 2023.

[Kocmi and Federmann, 2023b] Tom Kocmi and Christian Federmann. Large language models are state-of-the-art evaluators of translation quality. In *EAMT*, 2023.

[Kotonya *et al.*, 2023] Neema Kotonya, Saran Krishnasamy, Joel R. Tetreault, and Alejandro Jaimes. Little giants: Exploring the potential of small llms as evaluation metrics in summarization in the eval4nlp 2023 shared task. *CoRR*, abs/2311.00686, 2023.

[Kwon *et al.*, 2023] Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with pagedattention. In *SOSP*, 2023.

[Leiter *et al.*, 2023] Christoph Leiter, Juri Opitz, Daniel Deutsch, Yang Gao, Rotem Dror, and Steffen Eger. The eval4nlp 2023 shared task on prompting large language models as explainable metrics. *CoRR*, abs/2310.19792, 2023.

[Li *et al.*, 2022] Yujia Li, David Choi, Junyoung Chung, Nate Kushman, Julian Schrittwieser, Rémi Leblond, Tom Eccles, James Keeling, Felix Gimeno, Agustin Dal

Lago, Thomas Hubert, Peter Choy, Cyprien de Masson d'Autume, Igor Babuschkin, Xinyun Chen, Po-Sen Huang, Johannes Welbl, Sven Gowal, Alexey Cherepanov, James Molloy, Daniel J. Mankowitz, Esme Sutherland Robson, Pushmeet Kohli, Nando de Freitas, Koray Kavukcuoglu, and Oriol Vinyals. Competition-level code generation with alphacode. *Science*, 378(6624):1092–1097, 2022.

[Li *et al.*, 2023a] Junlong Li, Shichao Sun, Weizhe Yuan, Run-Ze Fan, Hai Zhao, and Pengfei Liu. Generative judge for evaluating alignment. *CoRR*, abs/2310.05470, 2023.

[Li *et al.*, 2023b] Qintong Li, Leyang Cui, Lingpeng Kong, and Wei Bi. Collaborative evaluation: Exploring the synergy of large language models and humans for open-ended generation evaluation. *CoRR*, abs/2310.19740, 2023.

[Li *et al.*, 2023c] Ruosen Li, Teerth Patel, and Xinya Du. PRD: peer rank and discussion improve large language model based evaluations. *CoRR*, abs/2307.02762, 2023.

[Li *et al.*, 2023d] Zongjie Li, Chaozheng Wang, Pingchuan Ma, Daoyuan Wu, Shuai Wang, Cuiyun Gao, and Yang Liu. Split and merge: Aligning position biases in large language model based evaluators. *CoRR*, abs/2310.01432, 2023.

[Lin and Chen, 2023] Yen-Ting Lin and Yun-Nung Chen. Llm-eval: Unified multi-dimensional automatic evaluation for open-domain conversations with large language models. In *NLP4ConvAI 2023*, 2023.

[Lin, 2004] Chin-Yew Lin. Rouge: A package for automatic evaluation of summaries. In *Text summarization branches out*, 2004.

[Liu *et al.*, 2023a] Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. G-eval: NLG evaluation using gpt-4 with better human alignment. In *EMNLP*, 2023.

[Liu *et al.*, 2023b] Yixin Liu, Alexander R. Fabbri, Jiawen Chen, Yilun Zhao, Simeng Han, Shafiq Joty, Pengfei Liu, Dragomir Radev, Chien-Sheng Wu, and Arman Cohan. Benchmarking generation and evaluation capabilities of large language models for instruction controllable summarization. *CoRR*, abs/2311.09184, 2023.

[Liu *et al.*, 2023c] Yixin Liu, Alexander R. Fabbri, Pengfei Liu, Dragomir Radev, and Arman Cohan. On learning to summarize with large language models as references. *CoRR*, abs/2305.14239, 2023.

[Liu *et al.*, 2023d] Yongkang Liu, Shi Feng, Daling Wang, Yifei Zhang, and Hinrich Schütze. Evaluate what you can't evaluate: Unassessable generated responses quality. *CoRR*, abs/2305.14658, 2023.

[Liu *et al.*, 2023e] Yuxuan Liu, Tianchi Yang, Shaohan Huang, Zihan Zhang, Haizhen Huang, Furu Wei, Weiwei Deng, Feng Sun, and Qi Zhang. Calibrating llm-based evaluator. *CoRR*, abs/2309.13308, 2023.

[Liusie *et al.*, 2023] Adian Liusie, Potsawee Manakul, and Mark J. F. Gales. Llm comparative assessment: Zero-shot nlg evaluation through pairwise comparisons using large language models. *Computing Research Repository*, arxiv:2307.07889, 2023.

[Lu *et al.*, 2023] Qingyu Lu, Baopu Qiu, Liang Ding, Liping Xie, and Dacheng Tao. Error analysis prompting enables human-like translation evaluation in large language models: A case study on chatgpt. *CoRR*, abs/2303.13809, 2023.

[Luo *et al.*, 2023] Zheheng Luo, Qianqian Xie, and Sophia Ananiadou. Chatgpt as a factual inconsistency evaluator for abstractive text summarization. *CoRR*, abs/2303.15621, 2023.

[Manakul *et al.*, 2023] Potsawee Manakul, Adian Liusie, and Mark J. F. Gales. Selfcheckgpt: Zero-resource black-box hallucination detection for generative large language models. In *EMNLP*, 2023.

[Mendonça *et al.*, 2023] John Mendonça, Patrícia Pereira, João Paulo Carvalho, Alon Lavie, and Isabel Trancoso. Simple LLM prompting is state-of-the-art for robust and multilingual dialogue evaluation. In *Proceedings of The Eleventh Dialog System Technology Challenge*, 2023.

[Menick *et al.*, 2022] Jacob Menick, Maja Trebacz, Vladimir Mikulik, John Aslanides, H. Francis Song, Martin J. Chadwick, Mia Glaese, Susannah Young, Lucy Campbell-Gillingham, Geoffrey Irving, and Nat McAleese. Teaching language models to support answers with verified quotes. *CoRR*, abs/2203.11147, 2022.

[Naismith *et al.*, 2023] Ben Naismith, Phoebe Mulcaire, and Jill Burstein. Automated evaluation of written discourse coherence using GPT-4. In *BEA@ACL*, 2023.

[Nakano *et al.*, 2021] Reiichiro Nakano, Jacob Hilton, Suchir Balaji, Jeff Wu, Long Ouyang, Christina Kim, Christopher Hesse, Shantanu Jain, Vineet Kosaraju, William Saunders, Xu Jiang, Karl Cobbe, Tyna Eloundou, Gretchen Krueger, Kevin Button, Matthew Knight, Benjamin Chess, and John Schulman. Webgpt: Browser-assisted question-answering with human feedback. *CoRR*, abs/2112.09332, 2021.

[Ostyakova *et al.*, 2023] Lidiia Ostyakova, Veronika Smilga, Kseniia Petukhova, Maria Molchanova, and Daniel Kornev. ChatGPT vs. crowdsourcing vs. experts: Annotating open-domain conversations with speech functions. In Svetlana Stoyanchev, Shafiq Joty, David Schlangen, Ondrej Dusek, Casey Kennington, and Malihe Alikhani, editors, *Proceedings of the 24th Annual Meeting of the Special Interest Group on Discourse and Dialogue*, pages 242–254, Prague, Czechia, September 2023. Association for Computational Linguistics.

[Ouyang *et al.*, 2022] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul F. Christiano, Jan Leike, and Ryan Lowe. Training language models to follow instructions with human feedback. In *NeurIPS*, 2022.

[Papineni *et al.*, 2002] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. Bleu: a method for automatic evaluation of machine translation. In *ACL*, 2002.

[Rastogi *et al.*, 2023] Charvi Rastogi, Marco Túlio Ribeiro, Nicholas King, Harsha Nori, and Saleema Amershi. Supporting human-ai collaboration in auditing llms with llms. In *AIES*, 2023.

[Ribeiro and Lundberg, 2022] Marco Túlio Ribeiro and Scott M. Lundberg. Adaptive testing and debugging of NLP models. In *ACL (1)*, 2022.

[Saha *et al.*, 2023] Swarnadeep Saha, Omer Levy, Asli Celikyilmaz, Mohit Bansal, Jason Weston, and Xian Li. Branch-solve-merge improves large language model evaluation and generation. *CoRR*, abs/2310.15123, 2023.

[Saunders *et al.*, 2022] William Saunders, Catherine Yeh, Jeff Wu, Steven Bills, Long Ouyang, Jonathan Ward, and Jan Leike. Self-critiquing models for assisting human evaluators. *CoRR*, abs/2206.05802, 2022.

[Shen *et al.*, 2023] Chenhui Shen, Liying Cheng, Xuan-Phi Nguyen, Yang You, and Lidong Bing. Large language models are not yet human-level evaluators for abstractive summarization. In *EMNLP (Findings)*, 2023.

[Shu *et al.*, 2023] Lei Shu, Nevan Wichers, Liangchen Luo, Yun Zhu, Yinxiao Liu, Jindong Chen, and Lei Meng. Fusion-eval: Integrating evaluators with llms. *CoRR*, abs/2311.09204, 2023.

[Sulem *et al.*, 2018] Elior Sulem, Omri Abend, and Ari Rappoport. BLEU is not suitable for the evaluation of text simplification. In *EMNLP*, 2018.

[Sun *et al.*, 2022] Tianxiang Sun, Junliang He, Xipeng Qiu, and Xuanjing Huang. Bertscore is unfair: On social bias in language model-based metrics for text generation. In *EMNLP*, 2022.

[Törnberg, 2023] Petter Törnberg. Chatgpt-4 outperforms experts and crowd workers in annotating political twitter messages with zero-shot learning. *CoRR*, abs/2304.06588, 2023.

[Touvron *et al.*, 2023] Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, Dan Bikel, Lukas Blecher, Cristian Canton-Ferrer, Moya Chen, Guillem Cucurull, David Esiobu, Jude Fernandes, Jeremy Fu, Wenyin Fu, Brian Fuller, Cynthia Gao, Vedanuj Goswami, Naman Goyal, Anthony Hartshorn, Saghar Hosseini, Rui Hou, Hakan Inan, Marcin Kardas, Viktor Kerkez, Madian Khabsa, Isabel Kloumann, Artem Korenev, Punit Singh Koura, Marie-Anne Lachaux, Thibaut Lavril, Jenya Lee, Diana Liskovich, Yinghai Lu, Yuning Mao, Xavier Martinet, Todor Mihaylov, Pushkar Mishra, Igor Molybog, Yixin Nie, Andrew Poulton, Jeremy Reizenstein, Rashi Rungta, Kalyan Saladi, Alan Schelten, Ruan Silva, Eric Michael Smith, Ranjan Subramanian, Xiaoqing Ellen Tan, Binh Tang, Ross Taylor, Adina Williams, Jian Xiang Kuan, Puxin Xu, Zheng Yan, Iliyan Zarov, Yuchen Zhang, Angela Fan, Melanie Kambadur, Sharan Narang, Aurélien Rodriguez, Robert Stojnic, Sergey Edunov, and Thomas Scialom. Llama 2: Open foundation and fine-tuned chat models. *CoRR*, abs/2307.09288, 2023.

[van der Lee *et al.*, 2021] Chris van der Lee, Albert Gatt, Emiel van Miltenburg, and Emiel Krahmer. Human evaluation of automatically generated text: Current trends and best practice guidelines. *Comput. Speech Lang.*, 67:101151, 2021.

[Varshney *et al.*, 2023] Neeraj Varshney, Wenlin Yao, Hongming Zhang, Jianshu Chen, and Dong Yu. A stitch in time saves nine: Detecting and mitigating hallucinations of llms by validating low-confidence generation. *CoRR*, abs/2307.03987, 2023.

[Wang *et al.*, 2023a] Jiaan Wang, Yunlong Liang, Fandong Meng, Haoxiang Shi, Zhixu Li, Jinan Xu, Jianfeng Qu, and Jie Zhou. Is ChatGPT a good NLG evaluator? a preliminary study. In *Proceedings of the 4th New Frontiers in Summarization Workshop*, 2023.

[Wang *et al.*, 2023b] Peiyi Wang, Lei Li, Liang Chen, Dawei Zhu, Binghuai Lin, Yunbo Cao, Qi Liu, Tianyu Liu, and Zhifang Sui. Large language models are not fair evaluators. *CoRR*, abs/2305.17926, 2023.

[Wang *et al.*, 2023c] Tianlu Wang, Ping Yu, Xiaoqing Ellen Tan, Sean O'Brien, Ramakanth Pasunuru, Jane Dwivedi-Yu, Olga Golovneva, Luke Zettlemoyer, Maryam Fazel-Zarandi, and Asli Celikyilmaz. Shepherd: A critic for language model generation. *CoRR*, abs/2308.04592, 2023.

[Wang *et al.*, 2023d] Yaqing Wang, Jiepu Jiang, Mingyang Zhang, Cheng Li, Yi Liang, Qiaozhu Mei, and Michael Bendersky. Automated evaluation of personalized text generation using large language models. *CoRR*, abs/2310.11593, 2023.

[Wang *et al.*, 2023e] Yidong Wang, Zhuohao Yu, Zhengran Zeng, Linyi Yang, Cunxiang Wang, Hao Chen, Chaoya Jiang, Rui Xie, Jindong Wang, Xing Xie, Wei Ye, Shikun Zhang, and Yue Zhang. Pandalm: An automatic evaluation benchmark for LLM instruction tuning optimization. *CoRR*, abs/2306.05087, 2023.

[Wang *et al.*, 2023f] Zifan Wang, Kotaro Funakoshi, and Manabu Okumura. Automatic answerability evaluation for question generation. *CoRR*, abs/2309.12546, 2023.

[Wu and Aji, 2023] Minghao Wu and Alham Fikri Aji. Style over substance: Evaluation biases for large language models. *CoRR*, abs/2307.03025, 2023.

[Wu *et al.*, 2023] Ning Wu, Ming Gong, Linjun Shou, Shining Liang, and Daxin Jiang. Large language models are diverse role-players for summarization evaluation. In *NLPCC (1)*, 2023.

[Xie *et al.*, 2023] Zhuohan Xie, Miao Li, Trevor Cohn, and Jey Han Lau. Deltascore: Fine-grained story evaluation with perturbations. In *EMNLP (Findings)*, 2023.

[Xu *et al.*, 2023] Wenda Xu, Danqing Wang, Liangming Pan, Zhenqiao Song, Markus Freitag, William Wang, and

Lei Li. INSTRUCTSCORE: towards explainable text generation evaluation with automatic feedback. In *EMNLP*, 2023.

[Ye *et al.*, 2023a] Seonghyeon Ye, Doyoung Kim, Sungdong Kim, Hyeonbin Hwang, Seungone Kim, Yongrae Jo, James Thorne, Juho Kim, and Minjoon Seo. FLASK: fine-grained language model evaluation based on alignment skill sets. *CoRR*, abs/2307.10928, 2023.

[Ye *et al.*, 2023b] Xi Ye, Srinivasan Iyer, Asli Celikyilmaz, Veselin Stoyanov, Greg Durrett, and Ramakanth Pasunuru. Complementary explanations for effective in-context learning. In *ACL (Findings)*, 2023.

[Yin and Neubig, 2022] Kayo Yin and Graham Neubig. Interpreting language models with contrastive explanations. In *EMNLP*, 2022.

[Yuan *et al.*, 2021] Weizhe Yuan, Graham Neubig, and Pengfei Liu. Bartscore: Evaluating generated text as text generation. In *NeurIPS*, 2021.

[Yuan *et al.*, 2024] Peiwen Yuan, Shaoxiong Feng, Yiwei Li, Xinglin Wang, Boyuan Pan, Heda Wang, and Kan Li. Batcheval: Towards human-like text evaluation. *CoRR*, abs/2401.00437, 2024.

[Zhang *et al.*, 2020] Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. Bertscore: Evaluating text generation with BERT. In *ICLR*, 2020.

[Zhang *et al.*, 2021] Yangjun Zhang, Pengjie Ren, and Maarten de Rijke. A human-machine collaborative framework for evaluating malevolence in dialogues. In *ACL/IJCNLP (1)*, 2021.

[Zhang *et al.*, 2022] Susan Zhang, Stephen Roller, Naman Goyal, Mikel Artetxe, Moya Chen, Shuohui Chen, Christopher Dewan, Mona T. Diab, Xian Li, Xi Victoria Lin, Todor Mihaylov, Myle Ott, Sam Shleifer, Kurt Shuster, Daniel Simig, Punit Singh Koura, Anjali Sridhar, Tianlu Wang, and Luke Zettlemoyer. OPT: open pre-trained transformer language models. *CoRR*, abs/2205.01068, 2022.

[Zhang *et al.*, 2023a] Chen Zhang, Luis Fernando D'Haro, Yiming Chen, Malu Zhang, and Haizhou Li. A comprehensive analysis of the effectiveness of large language models as automatic dialogue evaluators. *CoRR*, abs/2312.15407, 2023.

[Zhang *et al.*, 2023b] Xinghua Zhang, Bowen Yu, Haiyang Yu, Yangyu Lv, Tingwen Liu, Fei Huang, Hongbo Xu, and Yongbin Li. Wider and deeper LLM networks are fairer LLM evaluators. *CoRR*, abs/2308.01862, 2023.

[Zheng *et al.*, 2023] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging llm-as-a-judge with mt-bench and chatbot arena. *CoRR*, abs/2306.05685, 2023.

[Zhu *et al.*, 2023] Lianghui Zhu, Xinggang Wang, and Xinlong Wang. Judgelm: Fine-tuned large language models are scalable judges. *CoRR*, abs/2310.17631, 2023.